

Supplementary Material for “Prompt Stability Matters: A Benchmark for Quantifying Prompt Informativeness and Stability in Text-to-Image Models”

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A Additional Details on Stability Modeling

A.1 Building metrics from Mutual Information

From Mutual Information to CLIP Classification Accuracy. Let $Y \in \mathcal{Y}$ denote the discrete user intent category and $X \in \mathcal{X}$ the generated image. The mutual information between user intent and generated images is defined as

$$I(Y; X) = H(Y) - H(Y | X),$$

where $H(\cdot)$ denotes Shannon entropy. Since the prior distribution over intent categories is fixed across prompts and models, $H(Y)$ can be treated as a constant. Therefore, maximizing $I(Y; X)$ is equivalent to minimizing the conditional entropy $H(Y | X)$, which quantifies the remaining stability about the user intent after observing the generated image. However, the true posterior distribution $p(Y | X)$ is intractable. To obtain a computable surrogate, we introduce a classifier $f : \mathcal{X} \rightarrow \mathcal{Y}$, instantiated by CLIP [9], and define the predicted label $\hat{Y} = f(X)$. Let the classification error probability be

$$P_e = \Pr(\hat{Y} \neq Y).$$

By Fano’s inequality, the conditional entropy is upper-bounded as

$$H(Y | X) \leq h(P_e) + P_e \log(|\mathcal{Y}| - 1), \quad (1)$$

where $h(\cdot)$ denotes the binary entropy function. Since the right-hand side is a monotonic function of P_e , reducing the classification error directly reduces an upper bound on $H(Y | X)$, thereby increasing a lower bound on the mutual information $I(Y; X)$.

In practice, we estimate P_e using the empirical classification accuracy of CLIP over generated images:

$$\text{Acc} = \Pr(\hat{Y} = Y) = 1 - P_e.$$

Consequently, CLIP classification accuracy serves as a monotonic surrogate for mutual information between user intent and generated images. Higher accuracy implies lower conditional entropy $H(Y | X)$ and thus stronger intent-image alignment from an information-theoretic perspective.

Accuracy Variance and Mutual Information Stability. While the mean classification accuracy reflects the expected reduction of uncertainty in Y given X , it does not capture the variability of intent alignment across multiple samples generated from the same prompt. To characterize this aspect, we analyze the standard deviation of classification accuracy from an information-theoretic perspective.

Define a Bernoulli random variable

$$Z(X) = \mathbf{1}[\hat{Y}(X) = Y],$$

where $\hat{Y}(X)$ is the CLIP-predicted category. The empirical mean and variance of Z correspond to the classification accuracy and its standard deviation, respectively:

$$\mu = \mathbb{E}[Z], \quad \text{Var}(Z) = \mu(1 - \mu).$$

From Equation 1, $H(Y | X)$ is upper-bounded by a monotonic function of the image-specific classification error probability $P_e(X)$. Since $Z(X) = 1 - P_e(X)$, the variance of Z directly reflects the variance of $P_e(X)$ and thus the variability of the entropy $H(Y | X)$ across generated images.

Therefore, the standard deviation of classification accuracy estimates the dispersion of entropy $H(Y | X)$. A low variance indicates that $H(Y | X)$ is concentrated, implying stable and consistent intent transmission, whereas a high variance suggests a multi-modal generation distribution in which different samples convey disparate or conflicting semantic interpretations of the user intent.

MLLM Alignment Score as a Variational Approximation of Mutual Information. While classification-based metrics rely on hard decoding of user intent categories, they provide only coarse-grained estimates of the conditional entropy $H(Y | X, P)$. To capture fine-grained semantic alignment between generated images and user intent descriptions, we employ a multi-modal large language model (MLLM) to produce a soft alignment score.

Let Y denote the corresponding intent description expressed in natural language. The entropy of interest is defined as

$$H(Y | X) = \mathbb{E}_{X, Y}[-\log p(Y | X)],$$

where $p(Y | X)$ is the true but intractable posterior distribution. We approximate this distribution using an MLLM, which induces a variational distribution $q_\theta(Y | X)$ over intent descriptions given an image.

The expected cross-entropy between the true distribution and its variational approximation admits the standard decomposition:

$$\mathbb{E}[-\log q_\theta(Y | X)] = H(Y | X) + \mathbb{E}[\text{KL}(p \| q_\theta)],$$

where $\text{KL}(\cdot \| \cdot)$ denotes the Kullback–Leibler divergence. Since the KL term is non-negative, the expected negative log-likelihood under q_θ provides an upper bound on the true entropy:

$$H(Y | X) \leq \mathbb{E}[-\log q_\theta(Y | X)].$$

In practice, the MLLM produces a scalar alignment score $s(X, P) \in [0, 1]$, which can be interpreted as a calibrated estimate of $q_\theta(Y | X, P)$. Consequently, the expected negative log-alignment score,

$$\mathbb{E}[-\log s(X)],$$

serves as a computable surrogate for $H(Y | X, P)$. Since the conditional entropy appears as the only non-constant term in the prompt-conditioned mutual information,

$$I(Y; X | P = p) = H(Y | P = p) - H(Y | X, P = p),$$

maximizing the MLLM alignment score directly corresponds to increasing a variational lower bound on the mutual information between user intent and generated images.

Therefore, unlike classification accuracy which provides a discrete, upper-bounded estimate via Fano's inequality, the MLLM alignment score offers a continuous and semantically expressive approximation of mutual information, capturing fine-grained intent alignment beyond categorical correctness.

A.2 Prompt Entropy and Information-Theoretic Derivation

Consider the Markov chain $Y \rightarrow P \rightarrow X$, where Y is user intent, P is the prompt, and X is the generated image. By the data-processing inequality:

$$I(Y; X) \leq I(Y; P),$$

indicating that the maximum achievable stability is constrained by the prompt information content.

Operationally, for a prompt $P = (w_1, \dots, w_T)$, we approximate entropy using a pretrained LM:

$$H(P) \approx -\frac{1}{T} \sum_{t=1}^T \log p_{\theta}(w_t | w_1, \dots, w_{t-1}).$$

Interpretation:

- *Low cross-entropy*: Predictable, concentrated prompt effectively conveys user intent, enhancing stability.
- *High cross-entropy*: Uncertain or dispersed prompt, less informative, reducing stability.

A.3 Derivation of Prompt Energy

In classical statistical mechanics, a system state $x_t^{(i)}$ follows a Boltzmann distribution:

$$p(x_t^{(i)}) = \frac{\exp(-E_t^{(i)} / k\tau)}{Z_t}.$$

An autoregressive LM with parameters θ defines the probability of token x_t as

$$p_{\theta}(x_t | x_{<t}) = \frac{\exp(z_t(x_t))}{\sum_{v \in \mathcal{V}} \exp(z_t(v))},$$

where $z_t(v)$ is the logit of token v .

Identifying logits with negative energies up to a normalization constant C_t :

$$z_t(v) = -\frac{1}{k\tau} E_t(v) + C_t.$$

Setting $k\tau = 1$ and $C_t = 0$ yields token-level energy

$$e_t := E_t(x_t) = -z_t(x_t),$$

and sequence-level prompt energy

$$E(x) = -\frac{1}{T} \sum_{t=1}^T z_t(x_t),$$

which measures the model's confidence in generating x . Lower $E(x)$ indicates familiar, well-represented concepts, whereas higher $E(x)$ indicates uncertain or out-of-distribution concepts.

Usage. Prompt energy complements entropy and end-to-end mutual information metrics, enabling a more complete characterization of text-to-image generation stability.

B More Experiment Results

B.1 Experimental Setup Implementation Details

Training Hyperparameters Settings. We trained our model with the following hyperparameters: a learning rate of 1×10^{-5} , a batch size of 2, gradient accumulation steps of 16, and a total of 3 training epochs. The checkpoint with the lowest training loss was selected as the final model.

During LoRA fine-tuning, all parameters of the base model were frozen, and only the LoRA parameters were updated, specifically for query and value [5, 14]. The LoRA hyperparameters were set as follows: rank $r = 8$, $\alpha = 16$, a dropout rate of 0.1, and no bias. Training was performed using bf16 mixed precision.

We employed the Adam optimizer with $\beta_1 = 0.9$, $\beta_2 = 0.999$, and a weight decay of 1×10^{-2} .

NoxEye Prompt Template. To ensure consistency in model evaluation, we adopt the NoxEye Prompt Template, which specifies a unified structure system instruction, presenting tasks, inputs and outputs. The template is organized into four components:

- *System Instruction*: defines the global behavior and constraints that guide the model throughout the interaction.
- *Instruction*: defines the task description or objective to be performed.
- *Input*: provides the contextual information or query required to complete the task.
- *Response*: represents the expected model-generated output.

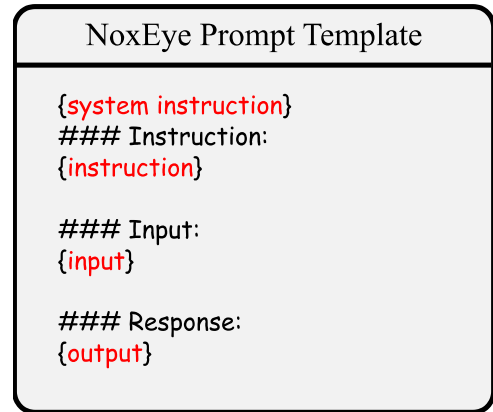


Figure 1: The prompt template of NoxEye.

Evaluation Settings. For evaluation on **AP Bench** and **GenEval**, the following hyperparameters were used: For SD2.1 [11] and CoMat2.1 [6], the number of sampling steps was set to 50, the CFG scale to 7.5, the

image size to default size. For other models, the hyperparameters are default parameters.

Preference Information Extractor Setting. We present the prompt to transform the LVL into the information extractor. The prompt for the preference information extractor in our model is illustrated in the Figure 2. As shown in Figure 3, the information of authentic prompts is significantly higher than that of prompts extracted by the preference information extractor, indicating that the latter produces prompts that are more concise and stable.

Preference Distribution Extractor Prompt

You are a professional AI image analyst specializing in analyzing Stable Diffusion generated images. Please analyze this image and generate a prompt that could have been used to create this image.

Requirements:

1. The generated prompt should be concise, accurate, and suitable for CLIP model understanding
2. Use English with comma-separated keyword format
3. Include the following elements (if applicable):
 - Subject description (people, objects, scenes)
 - Art style (e.g., realistic, anime, oil painting, digital art, etc.)
 - Quality descriptors (e.g., highly detailed, 8k, masterpiece, etc.)
 - Composition description (e.g., portrait, full body, close-up, etc.)
 - Lighting effects (e.g., soft lighting, dramatic lighting, etc.)
 - Color characteristics (e.g., vibrant colors, monochrome, etc.)
4. Avoid overly complex descriptions, keep the prompt practical
5. Sort by importance, with the most important keywords first

Please output the prompt directly without additional explanations.

Figure 2: Preference information extractor prompt template.

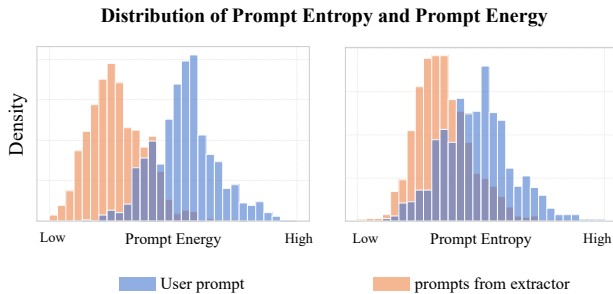


Figure 3: A visual comparison between authentic prompts and prompts from extractor.

B.2 Full Results of AP Bench

Tables 1, 2 and 3 report the full evaluation results on the Easy, Medium, and Hard levels of AP Bench under both novice and expert user settings. Overall, **NoxEye consistently achieves the best or near-best performance across all difficulty levels and backbones**, particularly in terms of alignment score. On the Easy and Hard levels, NoxEye also attains the highest accuracy mean while maintaining competitive or lower accuracy variance, indicating improved robustness. On the Medium level, NoxEye shows

clear advantages over prior methods for expert users and remains competitive for novice users. These results demonstrate that NoxEye generalizes well across user expertise, task difficulty, and diffusion backbones, outperforming existing prompt optimization approaches in both effectiveness and stability.

B.3 Full Results of GenEval

Table 6 reports the full evaluation results on the GenEval benchmark across multiple state-of-the-art text-to-image models. Overall, **NoxEye** demonstrates consistently strong and often superior performance across diverse model backbones and evaluation dimensions. On Stable Diffusion 2.1, NoxEye achieves the best results on *Single Object*, *Two Object*, *Counting*, and *Position*, leading to the highest overall score, which indicates improved compositional understanding and spatial grounding under complex prompts. For the more capable Stable Diffusion 3, while the original prompts yield the highest overall score, NoxEye remains highly competitive and attains near-best performance on fine-grained tasks such as *Counting* and *Two Object*, suggesting good generalization rather than over-specialization to a single model. Notably, on Flux.1[7] and PixArt- Σ [12], NoxEye consistently outperforms prior prompt optimization methods in terms of the overall metric, with clear gains on challenging attributes including multi-object composition, counting, and color reasoning. These results collectively indicate that NoxEye provides robust and model-agnostic improvements in text-image alignment, making it particularly effective for compositional and attribute-sensitive generation tasks, which aligns with the goals of controllable and reliable text-to-image synthesis.

B.4 Human Evaluation Details

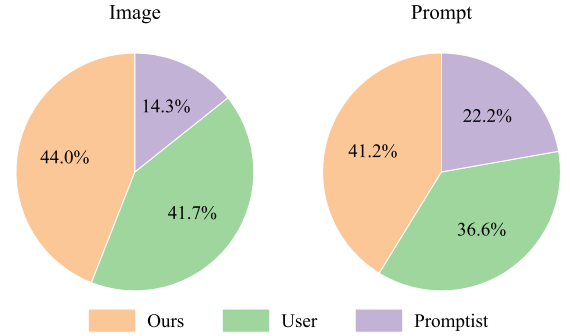
To complement quantitative evaluations with a human-centered perspective, we conducted a user study comparing our method with existing approaches. We first sampled a set of prompts at random and applied different optimization methods to obtain model-specific refined prompts, which were then used to generate images. A total of 20 trained experts with domain-specific professional backgrounds were recruited from diverse educational backgrounds. In each trial, participants were presented with either a pair of images or a pair of prompts and were asked to select the image they found more visually appealing or the prompt they preferred. As shown in Figure 4, participants most frequently selected images generated from prompts optimized by our method, indicating its superior effectiveness in aligning with human preference.

Table 1: Full Evaluation results on Easy level AP Bench. Bold values indicate best.

User type		Novice			Expert		
Method	Model	Acc Mean $\uparrow$	Acc Std $\downarrow$	Align. $\uparrow$	Acc Mean $\uparrow$	Acc Std $\downarrow$	Align. $\uparrow$
Promptist[4]	SD2.1[11]	0.3999	0.3742	0.6685	0.3450	0.3867	0.5237
	CoMat2.1[6]	0.4188	0.3754	0.6740	0.3651	0.3922	0.5317
	SD3[1]	0.4583	0.3719	0.7435	0.3847	0.3928	0.5759
	Flux.1[7]	0.4296	0.3694	0.7537	0.3778	0.3878	0.6003
	PixArt- Σ	0.4402	0.3708	0.7628	0.3260	0.3740	0.5954
	Inf[3]	0.4522	0.3758	0.7316	0.3483	0.3846	0.5655
	Show-O2[15]	0.3995	0.3684	0.7079	0.3011	0.3669	0.5372
PAG[16]	SD2.1[11]	0.3495	0.3774	0.5198	0.2754	0.3791	0.3462
	CoMat2.1[6]	0.3625	0.3791	0.5305	0.2819	0.3801	0.3546
	SD3[1]	0.3834	0.3834	0.5713	0.3120	0.3959	0.3875
	Flux.1[7]	0.3629	0.3737	0.5970	0.2890	0.3793	0.3916
	PixArt- Σ [12]	0.3594	0.3727	0.5996	0.2607	0.3653	0.4023
	Inf[3]	0.3740	0.3795	0.5738	0.2710	0.3816	0.3722
	Show-O2[15]	0.3352	0.3664	0.5507	0.2397	0.3565	0.3579
PAE[8]	SD2.1[11]	0.3659	0.3648	0.6596	0.3211	0.3686	0.6167
NoxEye (Ours)	SD2.1[11]	0.4268	0.3744	0.7047	0.3728	0.3832	0.6361
	CoMat2.1[6]	0.4439	0.3732	0.7113	0.3852	0.3865	0.6429
	SD3[1]	0.4760	0.3665	0.7670	0.4215	0.3851	0.6878
	Flux/1	0.4529	0.3670	0.7803	0.3958	0.3755	0.7018
	PixArt- Σ [12]	0.4651	0.3723	0.7890	0.3583	0.3681	0.7206
	Inf[3]	0.4627	0.3713	0.7659	0.3833	0.3794	0.6861
	Show-O2[15]	0.4108	0.3658	0.7281	0.3169	0.3540	0.6562

Table 2: Full Evaluation results on Medium level AP Bench. Bold values indicate best.

User type		Novice	Expert
Method	Model	Align. $\uparrow$	Align. $\uparrow$
Promptist [4]	SD2.1[11]	0.6416	0.5012
	CoMat2.1[6]	0.6302	0.5139
	SD3[1]	0.7035	0.5304
	Flux.1[7]	0.7071	0.5387
	PixArt- Σ [12]	0.7222	0.5541
	Inf[3]	0.7013	0.5221
	Show-O2[15]	0.6282	0.5053
PAG [16]	SD2.1[11]	0.5222	0.3248
	CoMat2.1[6]	0.5069	0.3293
	SD3[1]	0.5074	0.3448
	Flux.1[7]	0.5567	0.3611
	PixArt- Σ [12]	0.5521	0.3833
	Inf[3]	0.5586	0.3337
	Show-O2[15]	0.4772	0.3503
PAE [8]	SD2.1[11]	0.5885	0.6165
NoxEye (Ours)	SD2.1[11]	0.6524	0.6370
	CoMat2.1[6]	0.6443	0.6569
	SD3[1]	0.6992	0.7023
	Flux.1[7]	0.6958	0.7231
	PixArt- Σ [12]	0.7160	0.7113
	Inf[3]	0.6948	0.6617
	Show-O2[15]	0.6407	0.6536

**Figure 4: Human evaluation results. The result of NoxEye are preferred by human compared with the result of User Prompt and Promptist [4].**

B.5 Details of the Information Aligner Reward

To reduce hallucination tendencies and mitigate the squeezing effect in preference optimization [10, 13], we design the information aligner as an expert-guided composite reward function for GRPO-style preference optimization. Given an original user prompt P and a refined prompt P^* , we denote the reward as $r(P^*, P)$; for simplicity, the notation omits the generated image X produced from P , which is used during reward evaluation.

Specifically, the reward aggregates five complementary signals: (1) an LLM-based score evaluating whether the newly introduced

Table 3: Full Evaluation results on Hard level AP Bench. Bold values indicate best.

User type		Novice			Expert		
Method	Model	Acc Mean $\uparrow$	Acc Std $\downarrow$	Align. $\uparrow$	Acc Mean $\uparrow$	Acc Std $\downarrow$	Align. $\uparrow$
Promptist [4]	SD2.1[11]	0.1356	0.2848	0.6336	0.1424	0.2953	0.3786
	CoMat2.1[6]	0.1209	0.2667	0.6516	0.1209	0.2667	0.3826
	SD3[1]	0.1921	0.3242	0.6845	0.1774	0.3184	0.4179
	Flux.1[7]	0.1882	0.3263	0.7185	0.1389	0.2847	0.4437
	PixArt	0.1772	0.3279	0.6745	0.1191	0.2663	0.4504
	Inf[3]	0.1690	0.3083	0.6523	0.0991	0.2499	0.4392
	Show-O2[15]	0.1647	0.3141	0.6336	0.1206	0.2642	0.4009
PAG [16]	SD2.1[11]	0.1194	0.2603	0.3778	0.0918	0.2403	0.1900
	CoMat2.1[6]	0.1213	0.2670	0.3660	0.0978	0.2460	0.1886
	SD3[1]	0.1774	0.3125	0.3951	0.1137	0.2699	0.2174
	Flux.1[7]	0.1925	0.3299	0.4854	0.1021	0.2517	0.2140
	PixArt- Σ [12]	0.1520	0.2993	0.4437	0.0927	0.2449	0.2211
	Inf[3]	0.1645	0.3031	0.3993	0.0804	0.2337	0.2113
	Show-O2[15]	0.1499	0.2973	0.3850	0.0955	0.2478	0.1942
PAE [8]	SD2.1[11]	0.1232	0.2660	0.6576	0.1228	0.2729	0.4978
NoxEye (Ours)	SD2.1[11]	0.1498	0.3031	0.6840	0.1107	0.2560	0.5590
	CoMat2.1[6]	0.1216	0.2663	0.6833	0.1203	0.2667	0.5599
	SD3[1]	0.2011	0.3273	0.7220	0.1668	0.3143	0.5868
	Flux.1[7]	0.1902	0.3241	0.7718	0.1410	0.2874	0.6351
	PixArt- Σ [12]	0.1675	0.3153	0.6981	0.0952	0.2286	0.6662
	Inf[3]	0.1837	0.3320	0.6958	0.1140	0.2656	0.6157
	Show-O2[15]	0.1369	0.2912	0.6417	0.1037	0.2399	0.6011

Table 4: Evaluation results about CLIP Classification Accuracy (Mean $\uparrow$) on AP Bench. Bold values indicate best, and underlined values show second-best.

Method	SD2.1[11]	CoMat2.1[6]	SD3[1]	Flux.1[7]	PixArt- Σ [12]	Inf[3]	Show-O2[15]
Promptist [4]	0.3480	0.3655	0.3977	0.3751	0.3619	0.3731	0.3317
PAG [16]	0.2920	0.3019	0.3257	0.3060	0.2933	0.3033	0.2737
PAE [8]	0.3193	–	–	–	–	–	–
NoxEye (Mistral)	0.3184	0.3321	0.3868	0.3569	0.3496	0.3668	0.3213
NoxEye (Llama)	0.3681	0.3822	<u>0.4180</u>	<u>0.3940</u>	0.3829	<u>0.3921</u>	<u>0.3405</u>
NoxEye (Qwen)	0.3321	0.3438	0.3803	0.3592	0.3503	0.3640	0.3068

Table 5: Evaluation results about CLIP Classification Accuracy (standard deviation $\downarrow$) on AP Bench. Bold values indicate best, and underlined values show second-best.

Method	SD2.1[11]	CoMat2.1[6]	SD3[1]	Flux.1[7]	PixArt- Σ [12]	Inf[3]	Show-O2[15]
Promptist [4]	0.3783	0.3818	0.3834	0.3777	0.3758	0.3827	0.3681
PAG [16]	0.3726	<u>0.3749</u>	0.3849	0.3721	0.3673	0.3774	<u>0.3593</u>
PAE [8]	0.3639	–	–	–	–	–	–
NoxEye (Mistral)	<u>0.3690</u>	0.3718	0.3765	0.3655	<u>0.3687</u>	0.3764	0.3601
NoxEye (Llama)	0.3784	0.3803	0.3797	0.3744	0.3757	0.3795	0.3626
NoxEye (Qwen)	0.3744	0.3780	0.3817	0.3743	0.3730	0.3807	0.3560

Table 6: Full Evaluation results on GenEval [2]. Bold values indicate best.

Method	single Obj. ↑	Two Obj. ↑	Counting ↑	Colors ↑	Position ↑	Color Attri ↑	Overall
Stable Diffusion 2.1 [11]							
Original prompts	96.56%	50.76%	40.00%	84.31%	7.00%	15.00%	0.48938
Promptist [4]	96.25%	47.98%	41.25%	81.38%	7.25%	16.25%	0.48394
PAG [16]	94.69%	45.45%	34.06%	74.73%	4.50%	11.50%	0.44156
PAE [8]	85.62%	39.14%	23.12%	68.09%	6.50%	9.00%	0.38579
NoxEye (Ours)	97.50%	51.52%	43.44%	78.99%	11.00%	13.00%	0.4924
Stable Diffusion 3 [1]							
Original prompts	99.38%	86.87%	63.12%	87.50%	31.00%	61.50%	0.71561
Promptist [4]	99.06%	86.87%	59.69%	85.90%	33.25%	59.25%	0.7067
PAG [16]	98.75%	83.59%	52.50%	84.31%	22.00%	53.25%	0.65732
NoxEye (Ours)	99.38%	86.62%	61.88%	82.98%	29.25%	55.50%	0.69266
Flux.1 [schnell] [7]							
Original prompts	100.00%	86.87%	55.31%	74.47%	24.50%	45.50%	0.64442
Promptist [4]	96.25%	81.06%	50.31%	77.66%	29.75%	46.00%	0.63505
PAG [16]	97.50%	80.05%	46.56%	80.85%	22.75%	38.75%	0.61077
NoxEye (Ours)	99.06%	91.41%	58.13%	78.99%	29.25%	44.50%	0.6689
PixArt-Σ [12]							
Original prompts	99.38%	64.14%	43.75%	83.24%	12.50%	27.75%	0.55127
Promptist [4]	95.00%	58.08%	45.94%	79.52%	10.00%	25.00%	0.52257
PAG [16]	97.81%	63.38%	40.94%	81.12%	10.25%	21.00%	0.52417
NoxEye (Ours)	97.50%	67.42%	42.50%	85.64%	16.25%	27.50%	0.56135

content is useful and relevant rather than hallucinated ; (2) MLLM-based consistency scores to assess whether the generated result remains faithful to the original user intent in P ; (3) a SentenceTransformer-based information-increment score measuring the semantic difference between P^* and P ; (4) a CLIP-based text–image similarity score between P^* and X to measure prompt–image alignment; and (5) expert scores assigned to candidate refinements based on semantic faithfulness, informativeness, and hallucination suppression. In addition, expert judgments are used to supervise the model-based scoring process and filter out unreasonable or low-quality samples. By combining these signals, the reward encourages refinements that add informative details while preserving semantic faithfulness, thereby providing reliable preference signals for GRPO-style optimization.

As reflected by the alignment score in Table 7, the proposed reward is especially beneficial on the expert-hard subset of AP Bench, where prompts are generally longer, richer in detail, and harder to optimize due to ambiguous user intent. The consistent gains of GRPO over the SFT-only variant on most T2I backbones suggest that this expert-guided reward effectively steers the model toward more faithful and informative refinements, while reducing hallucination tendencies and alleviating the squeezing effect in challenging prompt-optimization scenarios.

Table 7: Impact of GRPO optimization. Bold values indicate best, and underlined values show second-best. NoxEye* refers to the model only training by SFT.

Method	SD2.1[11]	CoMat2.1[6]	SD3[1]	Flux.1[7]	PixArt-Σ[12]	Infl[3]	Show-o2[15]
Llama3	0.5349	0.5450	<u>0.5934</u>	0.6100	0.6411	0.6055	0.5649
NoxEye*(Ours)	<u>0.5445</u>	<u>0.5579</u>	0.6007	<u>0.6261</u>	<u>0.6635</u>	<u>0.6149</u>	<u>0.5987</u>
NoxEye (Ours)	0.5590	0.5599	0.5868	0.6351	0.6662	0.6157	0.6011

C More Quantitative Results

We present more visual results between the images generated with different prompts. As illustrated in the Figure 5, when a prompt simultaneously mentions both *fish* and *cat*, the generated images may omit the fish, indicating **informational sparsity**, where overly concise user prompts encourage LLM-based expansion to hallucinate unintended attributes and deviate from user intent. In addition, anomalous fish–cat amalgamations are observed, revealing **lexical perturbation and noise sensitivity**, in which minor word-level disturbances can mislead the text encoder and result in inconsistent or even contradictory generations.

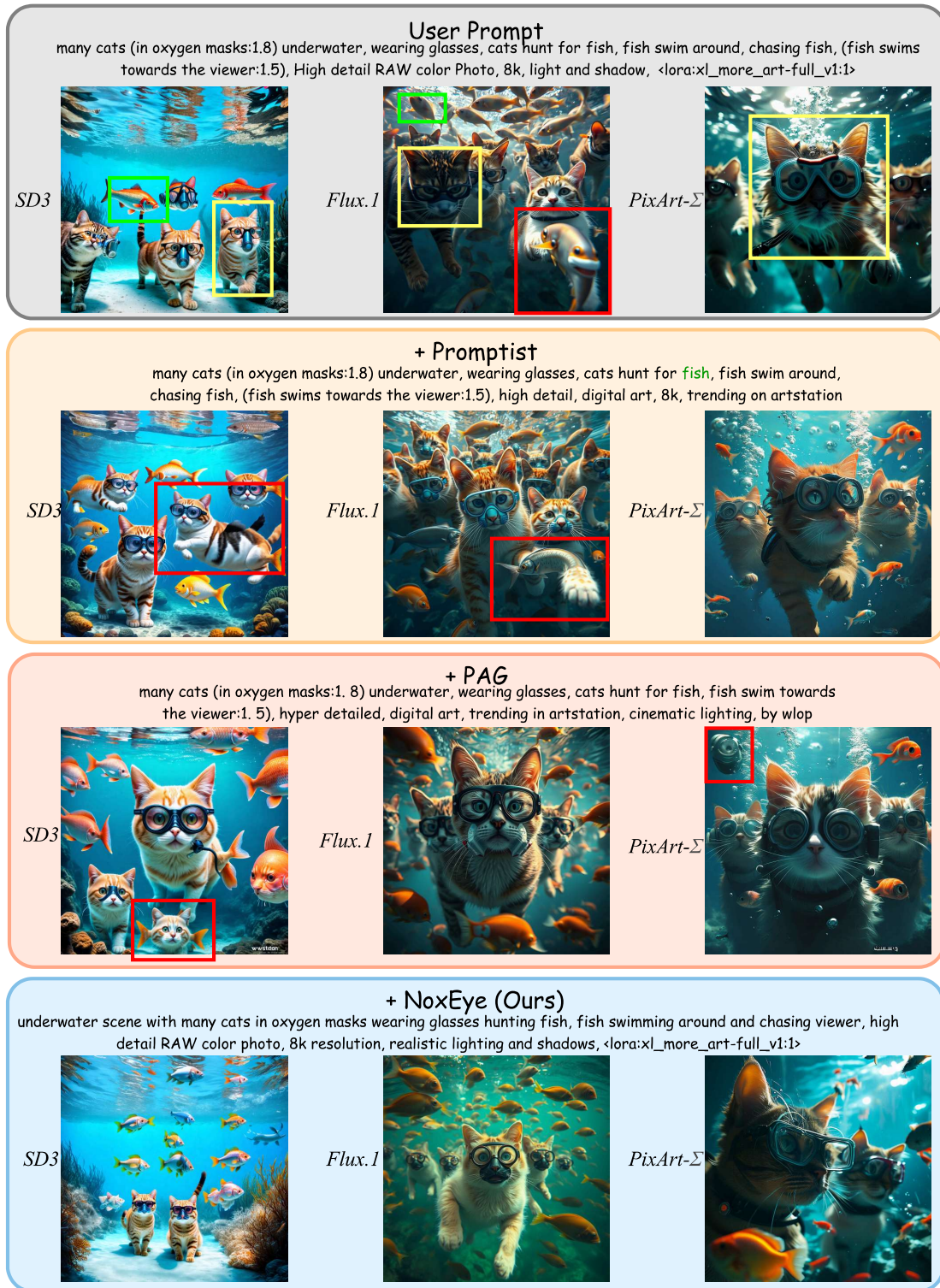


Figure 5: **Informational sparsity** stems from user prompts, while **lexical perturbation and noise sensitivity** arise in other prompt optimization approaches.

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